

DECODED EXPLAINED

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Contact lost with Drishti, world's first OPtoSAR satellite

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The Bengaluru-based space start-up GalaxEye has lost contact with Mission Drishti, the world's first OptoSAR satellite, as it encountered an anomaly following a geomagnetic solar storm, the company said on Tuesday.

Mission Drishti, which was launched in May, is also India's largest privately developed earth observation satellite.

"In the final stage of LE-OP (Launch and Early Orbit Phase), the maiden spacecraft encountered an anomaly following a geomagnetic solar storm. Initial root cause analysis indicates that radiation effects associated with the event likely impacted a critical onboard system," the company said.

It said subsequently, communication with the



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spacecraft became intermittent and was eventually lost. "While recovery efforts are ongoing, the likelihood of recovery currently appears low," it added.

Mission Drishti was launched on May 3 aboard a Falcon 9 by SpaceX from Vandenberg, California. The satellite integrated electro-optical (EO) and synthetic aperture radar (SAR) sensors into a single operational platform.

In UPSC syllabus

- GS Paper 3 (Mains) — **Science & Technology**: "Awareness in the fields of IT, Space, Computers, robotics, nano-technology, bio-technology..." — the exact syllabus line this fits under.
- GS Paper 3 (Mains) — also touches **Disaster Management** (space-weather/solar-storm hazard angle) and Indian Economy (private-sector investment, space economy share).
- Prelims — Science & Technology section, specifically Space Technology sub-topic.

What is an Earth Observation (EO) satellite?

An Earth Observation satellite is simply a **camera-carrying spacecraft** orbiting the Earth, constantly looking down and capturing images or data about the surface — farms, cities, borders, forests, disaster zones, border management, national security etc. Think of it as a CCTV camera for the entire planet, mounted hundreds of kilometres up in the sky instead of on a wall.

Two ways a satellite can see

There are two fundamentally different camera technologies used on Earth observation satellites. Every past Earth observation satellite had to pick ONE of these two systems.

1. Cartosat series primarily carries **optical sensors**,
2. while RISAT (**Radar (microwave pulses) Imaging Satellite**) carries SAR.

A user needing both all-weather coverage AND high-resolution optical detail traditionally needed data from two separate satellites, taken at two different times which creates gaps

Two ways a satellite can see

OPTICAL SENSOR

- Like a regular camera
- It needs sunlight
- Can't Works through clouds
- Ex- A tourist's camera — great pictures, but useless in fog or at night

SAR (Synthetic Aperture Radar)

- Sends down its own radar **(microwave) pulses and reads the echo bounced back**
- It can work in both day and night
- Ex - A bat's echolocation — works in total darkness, through obstacles

What is "OptoSAR"?

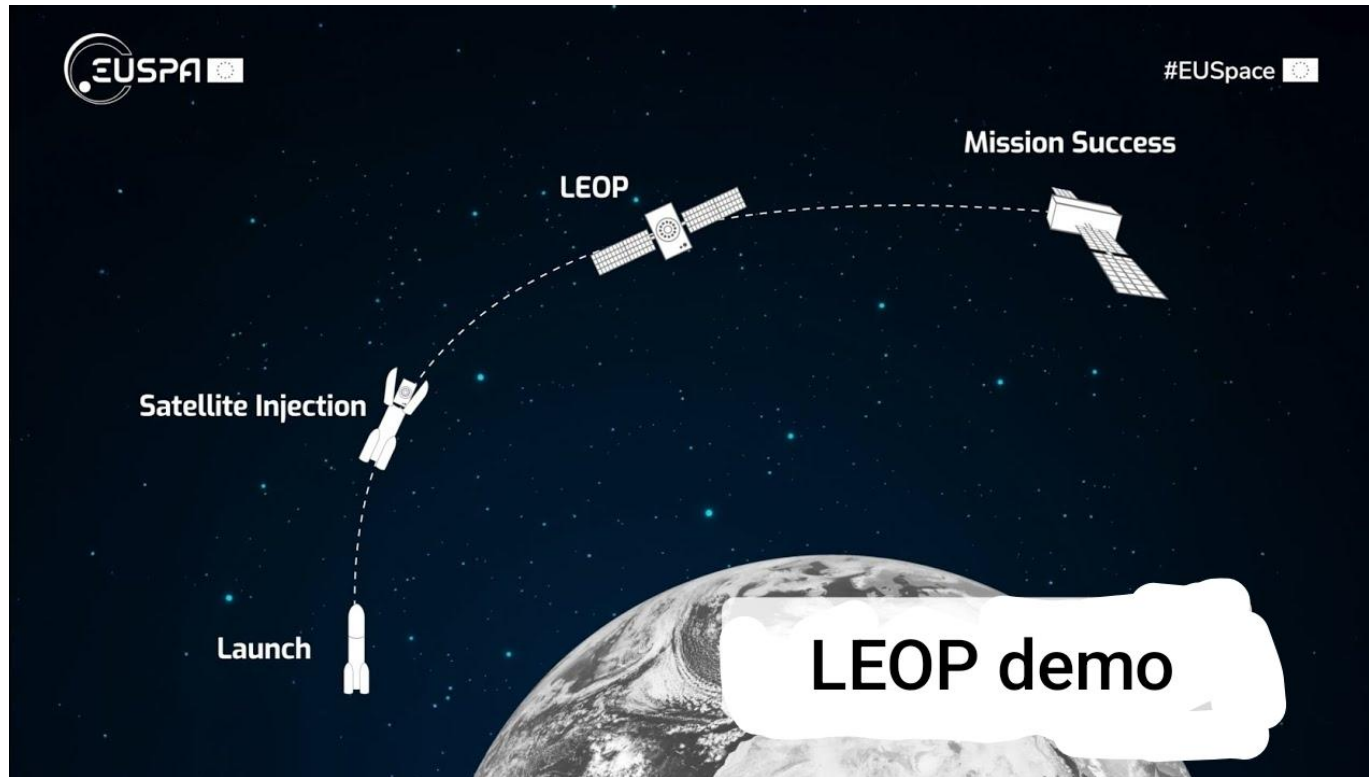
It is a satellite design concept where BOTH the **optical (EO) sensor** and the **SAR (radar) sensor** are fitted onto the same **single satellite platform**, working together .

Mission Drishti, built by the Bengaluru-based startup GalaxEye, was the **world's first satellite** to actually fly this combined **EO+SAR** design ,and it was also India's largest privately built Earth observation satellite

What is LEOP (Launch and Early Orbit Phase)?

LEOP is the most dangerous and delicate stretch of any satellite's life , the short window right after launch when the **satellite separates from the rocket, unfolds its solar panels and antennas, stabilises itself, and establishes first contact with ground control.**

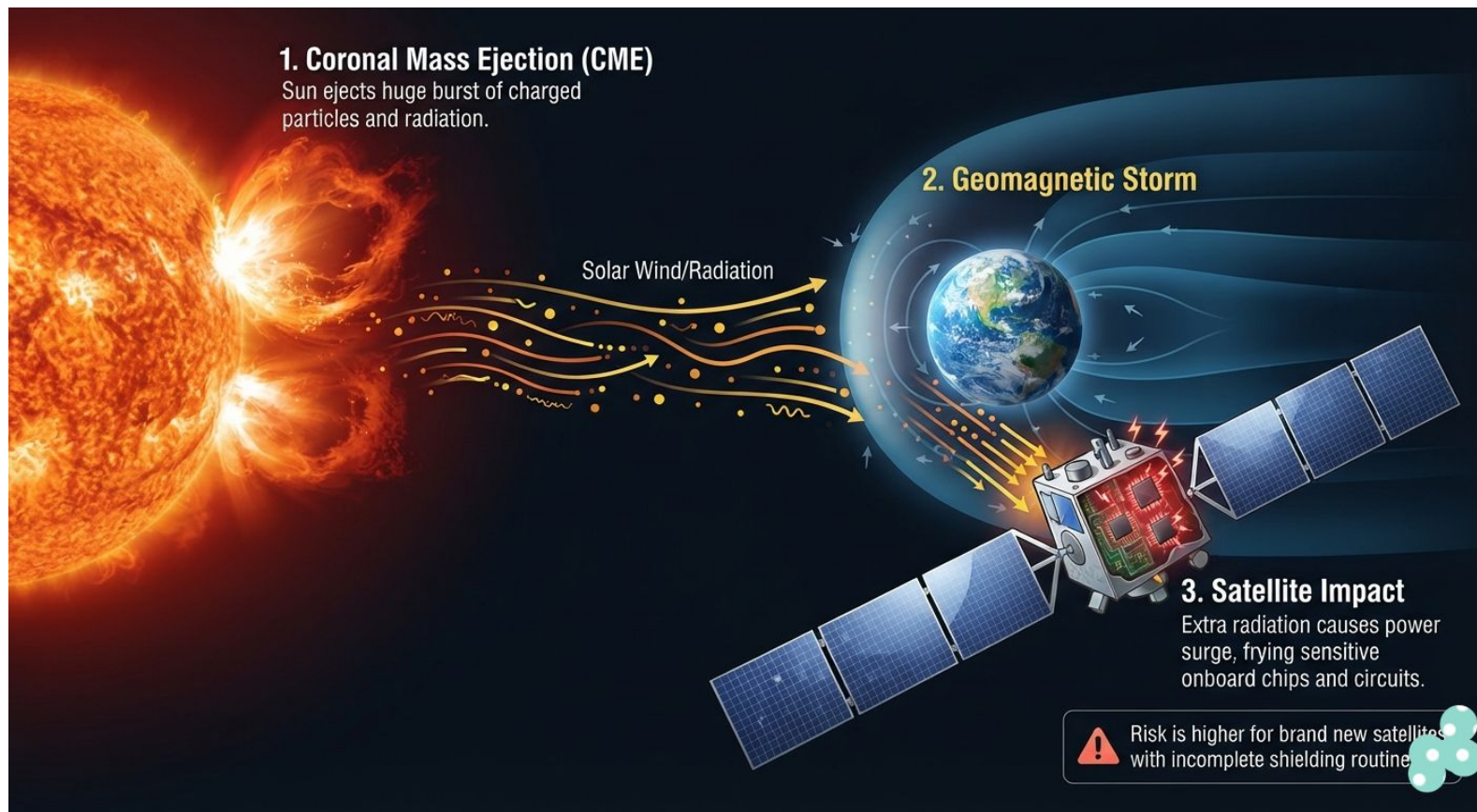
What is LEOP (Launch and Early Orbit Phase)?



What is a geomagnetic solar storm, and why does it break satellites?

The Sun periodically ejects huge bursts of **charged particles** into space (called a coronal mass ejection). When this reaches Earth, it disturbs our planet's magnetic field , this disturbance is called a **geomagnetic storm**.

For a satellite, this extra radiation is like a power surge hitting a home's electronics during a lightning storm ,it can fry sensitive onboard chips and circuits, especially in a satellite that is brand new and hasn't yet built up its full operational shielding routines.



PYQ

Consider the following statements with regard to involvement of private entities in India's space programme:

1. The Indian National Space Promotion and Authorisation Centre (IN-SPACe) is an autonomous agency formed to facilitate participation of private entities.
2. Agnikul Cosmos launched the world's first flight using 3D-printed rocket engine.
3. Skyroot Aerospace has developed liquid fuel for GSLV.

Which of the statements given above is/are correct?

(2026 Prelims)

PYQ

Which of the following are possible effects of a major solar storm (solar flare) on Earth?

GPS/navigation failure

Equatorial tsunamis

Power grid damage

Widespread intense auroras

Global forest fires

Satellite orbit disturbance

Polar flight radio interruption

(2022 Prelims)

MODEL QUESTION

India's private space sector is being positioned as a pillar of NewSpace growth, yet recent satellite mission setbacks raise questions about technological readiness. Discuss

NATO flaunts arms deal as Trump targets Europe allies

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Key 'defence forum' yields \$50 billion in deals: NATO official

Reuters
Ankara, July 7

NATO LEADERS unveiled arms deals worth tens of billions of dollars in Turkey on Tuesday, driving home the message that they are heeding US calls to spend more to defend Europe even as President Donald Trump said he felt let down and renewed his push to control Greenland.

Leaders were convening for a summit in the capital Ankara, hoping to project unity after another bruising year, in which the Iran war once again exposed cracks in the alliance.

In a meeting with President Tayyip Erdogan, Trump said he might have boycotted the NATO summit altogether had it not been for his warm relations with Turkish leader, and did not rule out further troop withdrawals from Europe. "Well, we're going to see. I was very disappointed with NATO," he said, singling out Britain, France, Germany and Italy for not doing enough to support US war on Iran.

Trump added "we weren't treated well" by the allies, even as he reiterated that he did not want or need their help. "Before I asked, they said they wouldn't be there, and we've invested trillions of dollars in NATO," Trump said.



US President Donald Trump with Turkish President Recep Tayyip Erdogan in Ankara on Tuesday. AP

NATO members have repeatedly tried to show Trump that they are stepping up. NATO Secretary General Mark Rutte said on Monday Europeans had made "staggering" increases in defence spending.

Before Trump's arrival, Rutte trumpeted a series of initiatives and deals at a defence industry forum, and called for a defence industry "revolution" across the alliance, warning over Russia's massive military spending as well as China, North Korea and Iran.

"We don't have the luxury of time. We need capabilities now to ensure we remain ready. The security situation demands it," Rutte said. "The hum of machinery must become a roar."

The deals, estimated to be worth at least \$50 billion ac-

cording to one NATO official, included European countries buying surveillance drones from US company Northrop Grumman, and NATO buying planes from Sweden's Saab.

Europe's defence sector has often been criticised as fragmented and saddled with red tape and rivalries between companies and countries. That has left Europe more reliant on purchases of US weapons.

Tensions within NATO, already strained over Ukraine and Trump's desire to wrest Greenland from fellow NATO member Denmark, have deepened since the US attacked Iran in February.

On Tuesday, he renewed his push to wrest Greenland from Denmark. "That should be controlled by the United States, not by Denmark," he said.

Trump: US to lift Turkey sanctions, sell F-35 fighters

Reuters
Ankara, July 7

PRESIDENT DONALD Trump on Tuesday announced that the US would lift sanctions on Turkey that were imposed in 2020 over Ankara's purchase of Russian missiles, and signaled a willingness to sell the NATO ally F-35 fighter jets, moves that would both be massive gestures to Turkey but may face legal hurdles.

In the first visit to Turkey by a US President in 11 years, Trump was welcomed by President Tayyip Erdogan at the airport with a lavish state ceremony.

Speaking to reporters at the Turkish presidential palace, Trump repeatedly praised Erdogan, saying he had "chemistry" with him and that the relationship with Ankara has never been better.

Turkey's 2019 acquisition of a Russian S-400 system and Washington's 2020 imposition of sanctions on a major Turkish defence company and removal of Turkey from the F-35 stealth fighter jet program have long been a sore point. "We're going to be taking the sanctions off," Trump said. "It's time. We don't want to sanction friends."

When asked if he had any concerns about Ankara still being in possession of the Russian S-400s, he said he did not.

In upsc syllabus

- GS Paper 2 (Mains) — International Relations: India's foreign policy vis-à-vis NATO, US-Europe relations, and their effect on India's strategic choices.
- Prelims — International Organisations

What is NATO?

NATO (North Atlantic Treaty Organization) is a **military alliance** formed in **1949** by the USA, Canada, and Western European countries, originally to counter Soviet power after World War II. Its headquarters is in **Brussels**, Belgium, and it now has **32 member countries**.

What is Article 5? (the core rule of NATO)

Article 5 is NATO's founding principle of "**collective defence**": an **armed attack against one member is treated as an attack against all members**, who must then assist the attacked country. This is the entire reason NATO exists as a credible deterrent , an enemy attacking one small member knows it is really picking a fight with all 32 countries, including the USA.

What is CAATSA ?

CAATSA (Countering America's Adversaries Through Sanctions Act) is a 2017 US law that allows Washington to **sanction any country that buys major defence hardware from Russia, Iran, or North Korea.**

PYQ

'The expansion and strengthening of NATO and a stronger US-Europe strategic partnership works well for India.' Give your opinion, with reasons and examples
(**2023 GS 2**)

Beyond three Cs, the new lexicon of India - Australia ties

Beyond three Cs, the new lexicon of India-Australia ties

During his visit to Australia in May 2023, when the Comprehensive Strategic Partnership between India and Australia was almost three years old, Prime Minister Narendra Modi had characterised India-Australia ties as having entered T20 mode. He also stressed that the relationship had gone beyond the conventional three Cs – of Commonwealth, Cricket and Curry – and the subsequent three Ds – Democracy, Diaspora and Dosti. As Mr. Modi undertakes his third visit to Australia this week, it would be instructive to look at some of the recent progress in two other important dimensions, the Ds of Development and Defence.

Expanding trade and investment

Growing bilateral economic cooperation has brought dividends like never before for both countries. All Indian exports to Australia now have duty free access under the Economic Cooperation and Trade Agreement (ECTA), benefiting textiles, pharmaceuticals, chemicals, engineering goods, gems and jewellery. Preferential access to 90% of Australia's trade value has facilitated Australian exports of critical minerals, resources, wool, avocados, and macadamia. There is the shared ambition to raise bilateral trade from \$33 billion in 2025 towards \$100 bn by 2030.

The two-way cumulative investment is approaching \$50 billion. Australia's AirTrunk recently announced plans to invest \$30 billion by 2030 to develop digital infrastructure and Artificial Intelligence (AI) ready data centre. Australian funds are increasingly active in India. At the same time, India is an important source of investment and technology. Indian investment in Australia's resources sector is well known. According to reports, Perdanian Chemicals & Fertilizers, founded by an Indian, is establishing Australia's largest urea plant in western Australia, with \$4.5 billion. This largest ever investment in Australia's fertilizer industry has also created millions of man hours of work in India where over 98% of the plant's modules are manufactured.

Defence represents perhaps the fastest growing sector of bilateral cooperation. In a clear signal of the importance of India as a reliable partner, Australia's Deputy Prime Minister and Defence Minister Richard Marles visited India in his first foreign tours in both the terms of the Albanese government. Before his most recent visit for the Annual Defence Ministers' Dialogue last month, Mr. Marles had hosted Defence



Gopal Baglay
Former High
Commissioner of
India to Australia

Minister Rajnath Singh in Australia last year, marking the first visit by an Indian Defence Minister to the country in 12 years. Regular leadership and other senior level exchanges involving all the three Services and participation in bilateral and multilateral military exercises such as AUSINDEX, Malabar and Talisman Sabre foster operational understanding and coordination, significant especially in the maritime domain.

There are growing opportunities for defence industry cooperation in cyber, AI, drone, and in view of India's expanding ship-building capabilities.

The 'E' factor

Beyond the Cs and the Ds, there are also enriching Es: Energy and Education. India-Australia Renewable Energy Partnership is implemented through a Solar Taskforce and a Green Hydrogen Task Force, and guided at the ministerial level.

India's unprecedented renewable energy targets present immense potential for clean energy cooperation across critical minerals and materials, manufacturing, laboratory research, commercial-scale deployment, industrial use, and solar rooftops.

Very recently, Australian media mentioned that arrangements for future Australian uranium exports to India might be finalised shortly. Were that to happen, India's ambitious civil nuclear programme will receive a significant boost, as would Australia's uranium exports.

The India-Australia education and skills partnership is steadily contributing to creation of an enduring base for future-ready youth, innovators and entrepreneurs. While more than one lakh Indian students are enrolled in Australia, world class and affordable education is now available in India itself through Indian campuses of a growing number of Australian universities. Joint research in priority areas such as advanced computing, energy, health care, space and defence promises to create both intellectual assets and networks. Specific visa programmes have created new avenues for educated Indian youth seeking employment opportunities in Australia and await better utilisation.

A growing partnership now spans trade, defence, energy, education and innovation. Australia's global leadership in imparting vocational skills is being tapped in areas such as solar rooftop and mining, including in Gujarat, Uttar Pradesh and Odisha. Practical solutions can pave way for meeting Australia's huge workforce

shortfall and spur its economy with Indian workers temporarily assigned to Australia.

Sport has emerged as a new priority. Focussed and broad based strategy can propel cooperation in sport education, training, medicine, goods, infrastructure and organisation of large-scale events, such as Commonwealth Games 2030 and Brisbane Olympics in 2032. With the growing Indian diaspora in Australia, now over ten lakh and rightly called the living bridge, Indian traditional sports such as kabaddi and kho kho are gaining popularity, even beyond the diaspora community.

Building multilateral partnerships

The upward trajectory of the bilateral relationship owes itself to a high degree of mutual trust, understanding and respect. Broad convergence of values and interests expands the format of cooperation to trilateral, such as India-Indonesia-Australia and India-France-Australia. The Australia-Canada-India Technology and Innovation Partnership, launched in November 2025, the India-Japan-Australia Supply Chain Resilience Initiative and a possible triad with the United Arab Emirates could play important roles in countering supply chain disruptions and market dominance and distortions in crucial areas such as critical minerals, rare earths, semiconductor and new technologies.

The shared vision of India and Australia for a free, open, safe, peaceful and prosperous Indo-Pacific manifests in their work through fora such as the Quad and the Indian Ocean Rim Association (IORA). The foreign visits this week by Mr. Modi and Australian Prime Minister Anthony Albanese bring out the geopolitical logic for ever-growing cooperation between India and Australia at regional and global levels. Their existing ties with Pacific Island Countries provide a good basis for leveraging their respective strengths to the benefit of the region in areas such as education, health, technology, fintech, capacity building and disaster relief.

Mr. Albanese never tires of mentioning his connection with India as a young backpacker. His understanding of the significance of a consequential relationship with today's India and close chemistry between the two leaders have set the perfect stage to progress bilateral ties in T20 mode but in the long format, and to multiple win-win outcomes for both.

The views expressed are personal

A growing
partnership now
spans trade,
defence, energy,
education and
innovation

How did India-Australia ties evolve

India-Australia Ties ,The Evolving Alphabet

Stage 1 — Three Cs

1. Commonwealth — shared British-colonial-era institutional membership
2. Cricket — shared sporting culture
3. cultural ties

India-Australia Ties ,The Evolving Alphabet

Stage 2 — four Ds

1. Democracy — shared political system
2. Diaspora — large Indian community in Australia
3. Development — economic/trade cooperation
4. Defence — security and military ties

India-Australia Ties ,The Evolving Alphabet

Stage 3 — Two Es (latest addition)

1. Energy — renewables, critical minerals, uranium
2. Education — student mobility, university partnerships

What is the Comprehensive Strategic Partnership (CSP)?

A Comprehensive Strategic Partnership is a formal upgrade in bilateral ties, signalling that two countries intend to **cooperate across nearly every domain** , trade, defence, education, technology etc , not just one or two areas.

India and Australia elevated their relationship to a CSP in the year 2020.

What is ECTA, and what is CECA?

- **ECTA - Economic Cooperation and Trade agreement**
- **What is it - A limited early-stage trade deal, in maximum one or two domain**
- **CECA - Comprehensive Economic Cooperation Agreement**
- **What is it - A more ambitious, broader trade pact that build on ECTA, multiple domains**

What is the Quad, and what is IORA?

- The **Quad** (Quadrilateral Security Dialogue) is an informal strategic grouping of India, Australia, Japan, and the USA, focused on keeping the Indo-Pacific region free, open, and rules-based
- **IORA** (Indian Ocean Rim Association) is a separate regional grouping of **countries bordering the Indian Ocean**, focused on trade, maritime safety, and the "blue economy" (economic activity linked to oceans, like fisheries and shipping) — both India and Australia are members.

How India Australia agreement contribute to bilateral trade and investment growth?

- All Indian exports now get **duty-free access to Australia** under ECTA, benefiting **textiles, pharmaceuticals, chemicals, engineering goods, gems and jewellery**
- Gave India preferential access to 90% of Australia's trade value, boosting Australian exports including **critical minerals**, resources, wool, avocados, and to India
- Shared ambition to raise bilateral trade from ~\$33 billion (2025) toward \$100 billion by 2030
- Indian-founded **Perdaman Chemicals & Fertilizers** is building Australia's largest urea plant, India's largest-ever investment in Australia's fertilizer industry

What potential do future Australian uranium exports hold for India's civil nuclear programme?

- uranium is the core fuel input for nuclear power generation
- **Diversifies** India's uranium sourcing beyond current **suppliers** like Russia, Kazakhstan, France
- **Strengthens long-term fuel security** for existing and upcoming nuclear reactors nationwide

Way forward

- Set **concrete timelines** for concluding CECA rather than repeated "acceleration" pledges without a fixed deadline
- **Build measurable delivery** benchmarks
- Deepen industrial cooperation to achieve the target of 100bn by 2030

India's cooperative sector can serve as a worker-centric model of development

Grand ambitions

India's cooperative sector can serve as a worker-centric model of development

In a hypercompetitive business models that dominate the economic lives of people have undesirable social consequences. The cooperative model offers an alternative, albeit with its own imperfections. India's Ministry of Cooperation, which completed five years on July 6, is a bold experiment in harnessing the potential of this approach. Traditionally confined largely to agriculture, cooperatives now have the opportunity and a requirement to expand into other sectors, particularly services. Cooperatives, by their very character, are relatively small-scale and fragmented; organising them, and connecting them to the broader economy that disproportionately rewards hyper-scalers is a balancing intervention – an economic, political and social imperative. The Ministry aims to transform the cooperative landscape by bringing policy coherence across agriculture, dairy, fisheries, banking, housing, consumer cooperatives, and exports. This requires collaboration with States and national federations, capacity building, wise use of digital technology, and market linkages. **Primary Agricultural Credit Societies (PACS) are the foundation of rural cooperatives.** Through a reformed legal framework, they have been empowered to undertake over 25 business activities, transforming them into multifaceted institutions delivering many economic services in rural India.

The Ministry has facilitated new national-level multi-State cooperative societies, expanding market access for members and strengthening cooperative value chains from production to global markets. Cooperative enterprises are being encouraged to grow and compete, and a National Cooperation Policy is in the making. But there are challenges amid the opportunities. Corruption and inefficiency eroded the potential of India's cooperative sector, and the fear of local communities and States losing control of cooperatives to a national-level mechanism needs to be assuaged. The task is in finding the sweet spot of consolidation and decentralisation; localisation and nationalisation; technology and human labour, for boosting the sector. Union Home Minister Amit Shah, who holds the portfolio, is pushing the cooperative sector to expand into production and marketing, beyond agricultural credit and input facilitation, which could help mitigate the sector's endemic crisis. A well-coordinated, yet federated, cooperative sector can offset the social, environmental and political costs of global capitalism, which is the default model of the economy. Government and business leaders worldwide are viewing cooperatives with renewed interest, as the pitfalls of hypercapitalism become pronounced. India can develop its cooperative sector as a global model.

In UPSC syllabus

- GS Paper 2 (Mains) — Governance: cooperative federalism, 97th Constitutional Amendment, Centre-State jurisdiction over cooperatives.
- GS Paper 3 (Mains) — Economy: agricultural credit, rural cooperative marketing, alternative economic models.
- Prelims — Polity (97th Amendment, Part IX-B) and Economy (PACS, Urban Cooperative Banks, MSCS Act).

What is a cooperative society?

A group of people come together, each contributing a little money or effort, to run something for everyone's shared benefit , not for one owner's profit, but for the members themselves.

Example -

- Think of a cooperative like a group of neighbours who pool money to buy groceries in bulk from a wholesaler, instead of buying individually from retail saler. The saved money then split between the members

What constitutional status do cooperatives have in India?

- **97th Constitutional Amendment Act, 2011** gave cooperatives formal constitutional recognition
- Fundamental Right — **Article 19(1)(c): right to form cooperative societies**
- DPSP added — **Article 43-B:** directs State to promote cooperatives
- New **Part IX-B** was created

Who Governs Cooperatives — Centre vs. States

- Cooperation is a **State subject** — Entry 32, State List (List II), Seventh Schedule
- Exception: **Parliament** can legislate for **Multi-State Cooperative Societies** (those operating across more than one state)

PYQ

UPSC Prelims 2021:

With reference to 'Urban Cooperative Banks' in India, consider the following statements:

1. They are supervised and regulated by local boards set up by the State Governments.
2. They can issue equity shares and preference shares.
3. They were brought under the purview of the Banking Regulation Act, 1949 through an Amendment in 1966.

THANK YOU